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United States Patent	12402494
Kind Code	B2
Date of Patent	August 26, 2025
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Organic light-emitting display device having reduced compensated step from electrode lines to prevent cracking of encapsulation layer

Abstract

An organic light-emitting display device includes a substrate having a display region and a peripheral region, a plurality of pixels on the substrate in the display region, a first wiring and a second wiring on the substrate in the peripheral region, An insulation layer on the first and second wirings, the insulation layer covering a top surface and a sidewall of each of the first and second wirings, and an encapsulation layer on the plurality of pixels and on the insulation layer.

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Appl. No.: 18/630503

Filed: April 09, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240260339 A1	Aug. 01, 2024

Foreign Application Priority Data

KR	10-2015-0183520	Dec. 22, 2015
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Related U.S. Application Data

continuation parent-doc US 17866858 20220718 US 12058900 child-doc US 18630503
continuation parent-doc US 16903638 20200617 US 11411060 20220809 child-doc US 17866858
continuation parent-doc US 15386526 20161221 US 10714553 20200714 child-doc US 16903638

Publication Classification

Int. Cl.: **H10K59/124** (20230101); **H10K50/844** (20230101); **H10K59/121** (20230101);
H10K59/131 (20230101); **H10K59/80** (20230101)

U.S. Cl.:

CPC **H10K59/124** (20230201); **H10K50/8445** (20230201); **H10K59/1213** (20230201);
H10K59/131 (20230201); **H10K59/8731** (20230201); H10K59/8051 (20230201);
H10K59/8052 (20230201)

Field of Classification Search

CPC: H10K (59/124); H10K (59/131); H10K (59/1213); H10K (50/8445); H10K (59/8731);
H10K (59/8051); H10K (59/8052)

USPC: 257/40; 257/59; 257/99

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This is a continuation application of U.S. patent application Ser. No. 17/866,858 filed Jul. 18, 2022, which issues as U.S. Pat. No. 12,058,900 on Aug. 6, 2024, the disclosure of which is incorporated herein by reference in its entirety. U.S. patent application Ser. No. 17/866,858 is a continuation application of U.S. patent application Ser. No. 16/903,638 filed Jun. 17, 2020, now U.S. Pat. No. 11,411,060, issued Aug. 9, 2022, the disclosure of which is incorporated herein by reference in its entirety. U.S. patent application Ser. No. 16/903,638 is a continuation application of U.S. patent application Ser. No. 15/386,526, filed Dec. 21, 2016, now U.S. Pat. No. 10,714,553, issued Jul. 14, 2020. U.S. patent application Ser. No. 15/386,526 claims priority benefit of Korean Patent Application No. 10-2015-0183520, filed on Dec. 22, 2015 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety for all purposes.

BACKGROUND

1. Field

(1) Example embodiments relate to display devices. More particularly, example embodiments relate to organic light-emitting display devices including a compensation layer that compensates a stepped region derived from wirings.

2. Description of the Related Art

(2) An organic light-emitting display device may have an organic light-emitting structure including a hole injection layer, an electron injection layer, and an organic light-emitting layer formed therebetween. In the organic light-emitting display device, light may be generated as excitons, that are the combination of holes injected from the hole injection layer and electrons injected from the electron injection layer, fall from an excited state to a ground state. The organic light-emitting display device may not require a separate light source, and thus the organic light-emitting display device may have a small thickness and a light weight, as well as low power consumption. Furthermore, the organic light-emitting display device may have a wide viewing angle, a high contrast, and a high response speed, etc.

(3) However, the organic light-emitting structure may be easily degraded by vapor or oxygen from the outside. Therefore, an encapsulation member may be required to protect the organic light-emitting structure from the vapor or the oxygen. Recently, an encapsulation layer has been used as the encapsulation member for the organic light-emitting structure to make the organic light-emitting structure thin and flexible.

SUMMARY

(4) According to an aspect of example embodiments, an organic light-emitting display device may include a substrate having a display region and a peripheral region, a plurality of pixels disposed on the substrate in the display region, a first wiring and a second wiring disposed on the substrate in the peripheral region, a compensation layer disposed on the first and second wirings, and an encapsulation layer disposed on the plurality of pixels and on the compensation layer. The compensation layer may surround a top surface and a sidewall of each of the first and second wirings.

(5) In example embodiments, the compensation layer may compensate a stepped region derived from the first and second wirings.

(6) In example embodiments, the first wiring may supply a driving voltage to the plurality of pixels, and the second wiring may supply a common voltage to the plurality of pixels.

(7) In example embodiments, each of the plurality of pixels may include a pixel circuit having an active pattern, a gate electrode, an insulation interlayer, a source electrode and a drain electrode, and an organic light-emitting structure having a pixel electrode, an organic light-emitting layer and a common electrode. The organic light-emitting structure may be electrically connected to the pixel circuit.

(8) In example embodiments, the first and second wirings, and the gate electrode may be disposed

at substantially the same level over the substrate. The compensation layer and the insulation interlayer may be disposed at substantially the same level over the substrate.

(9) In example embodiments, each of the plurality of pixels may further include an insulation layer disposed between the source and the drain electrodes, and the pixel electrode, and a pixel defining layer disposed on the insulation layer. The pixel defining layer may expose a portion of the pixel electrode.

(10) In example embodiments, the encapsulation layer may include at least one organic layer and at least one inorganic layer.

(11) In some example embodiments, the encapsulation layer may have a structure in which a first inorganic layer, an organic layer, and a second inorganic layer are stacked.

(12) According to another aspect of example embodiments, an organic light-emitting display device may include a substrate having a display region and a peripheral region, a plurality of pixels disposed on the substrate in the display region, a first wiring and a second wiring disposed on the substrate in a peripheral region, an interwiring insulation layer surrounding a top surface and a sidewall of each of the first and second wirings, a third wiring disposed on the interwiring insulation layer, the third wiring being electrically connected to the first wiring, a fourth wiring disposed on the interwiring insulation layer, the fourth wiring being electrically connected to the second wiring, a compensation layer disposed on the third and fourth wirings, and an encapsulation layer disposed on the plurality of pixels and on the compensation layer. The compensation layer may surround a top surface and a sidewall of each of the third and fourth wirings.

(13) In example embodiments, the compensation layer may compensate a stepped region derived from the third and fourth wirings.

(14) In example embodiments, the first and the third wirings may supply a driving voltage to the plurality of pixels, and the second and the fourth wirings may supply a common voltage to the plurality of pixels.

(15) In example embodiments, the interwiring insulation layer may have a first contact hole exposing the first wiring and a second contact hole exposing the second wiring. The third wiring may fill the first contact hole to be contacted to the first wiring, and the fourth wiring may fill the second contact hole to be contacted to the second wiring.

(16) In example embodiments, each of the plurality of pixels may include a pixel circuit having an active pattern, a first gate electrode, a gate insulation layer, a second gate electrode, an insulation interlayer, a source electrode and a drain electrode, and an organic light-emitting structure having a pixel electrode, an organic light-emitting layer and a common electrode. The organic light-emitting structure may be electrically connected to the pixel circuit.

(17) In example embodiments, the first and second wirings, and the first gate electrode may be disposed at substantially the same level over the substrate. The third and fourth wirings, and the second gate electrode may be disposed at substantially the same level over the substrate.

(18) In example embodiments, the compensation layer and the insulation interlayer may be disposed at substantially the same level over the substrate. The interwiring insulation layer and the gate insulation layer may be disposed at substantially the same level over the substrate.

(19) In example embodiments, each of the plurality of pixels may further include an insulation layer disposed between the source and the drain electrodes, and the pixel electrode, and a pixel defining layer disposed on the insulation layer. The pixel defining layer may expose a portion of the pixel electrode.

(20) In example embodiments, the encapsulation layer may include at least one organic layer and at least one inorganic layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Features will become apparent to those of ordinary skill in the art by describing in detail exemplary embodiments with reference to the attached drawings, in which:
- (2) FIG. 1 illustrates a plan view of an organic light-emitting display device in accordance with example embodiments.
- (3) FIG. 2 illustrates a circuit diagram of a pixel of the organic light-emitting display device in FIG. 1.
- (4) FIG. 3 illustrates a plan view of a portion 'X' of the organic light-emitting display device in FIG. 1.
- (5) FIG. 4 illustrates a cross-sectional view along line I-I' of FIG. 1.
- (6) FIG. 5 illustrates a cross-sectional view along line I-I' of FIG. 1 in accordance with other example embodiments.

DETAILED DESCRIPTION

- (7) Example embodiments will now be described more fully hereinafter with reference to the accompanying drawings; however, they may be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey exemplary implementations to those skilled in the art.
- (8) In the drawing figures, the dimensions of layers and regions may be exaggerated for clarity of illustration. It will also be understood that when a layer or element is referred to as being "on" another layer or substrate, it can be directly on the other layer or substrate, or intervening layers may also be present. In addition, it will also be understood that when a layer is referred to as being "between" two layers, it can be the only layer between the two layers, or one or more intervening layers may also be present. Like reference numerals refer to like elements throughout.
- (9) Hereinafter, organic light-emitting display devices in accordance with example embodiments will be explained in detail with reference to the accompanying drawings.
- (10) FIG. 1 is a plan view illustrating an organic light-emitting display device in accordance with example embodiments. FIG. 2 is a circuit diagram illustrating a pixel of the organic light-emitting display device in FIG. 1.
- (11) Referring to FIG. 1, an organic light-emitting display device **100** may include a plurality of pixels PX, a first wiring part **130**, a second wiring part **140**, an encapsulation layer **150**, a pad part **170**, an embedded circuit part **180**, etc. For example, the organic light-emitting display device **100** may have a display region **110** and a peripheral region **120**.
- (12) An image may be displayed by the pixels PX in the display region **110** of the organic light-emitting display device **100**. For example, the display region **110** may be substantially located on a center portion of the organic light-emitting display device **100**.
- (13) A plurality of lines **175**, **185** and **195** may be disposed in the display region **110**. The lines **175**, **185** and **195** may include a plurality of data lines **175** for transferring data signals to the pixels PX, a plurality of gate lines **185** for transferring gate signals (or scan signals) to the pixels PX, and a plurality of power voltage lines **195** for transferring a driving voltage ELVDD to the pixels PX, etc. The data lines **175** and the power voltage lines **195** may be substantially parallel to each other, and may extend along a first direction. The gate lines **185** may extend along a second direction substantially perpendicular to the first direction.
- (14) The plurality of pixels PX disposed in the display region **110** for displaying the image may be arranged as a substantial matrix structure.
- (15) Referring to FIG. 2, each pixel PX may include a pixel circuit and an organic light-emitting structure **330**. The pixel circuit may include a switching transistor **300**, a driving transistor **310**, a capacitor **320**, etc. FIG. 2 illustrates the pixel circuit having two transistors and one capacitor, however embodiments may not be limited thereto, e.g., the pixel circuit may include at least three

transistors and/or at least two capacitors.

(16) The switching transistor **300** may have a control electrode, an input electrode, and an output electrode. The control electrode may be connected to the gate line **185**, the input electrode may be connected to the data line **175**, and the output electrode may be connected to the driving transistor **310**. The switching transistor **300** may transfer the data signal from the data line **175** to the driving transistor **310** based on the gate signal from the gate line **185**.

(17) The driving transistor **310** may have a control electrode, an input electrode, and an output electrode. The control electrode may be connected to the switching transistor **300**, the input electrode may be connected to the power voltage line **195**, and the output electrode may be connected to the organic light-emitting structure **330**. The driving transistor **310** may transfer an output current I_D to the organic light-emitting structure **330** based on a voltage between the control electrode and the output electrode.

(18) The capacitor **320** may be connected between the control electrode and the input electrode of the driving transistor **310**. The capacitor **320** may charge the data signal applied to the control electrode of the driving transistor **310**, and the capacitor **320** may maintain the data signal after the switching transistor **300** is turned-off.

(19) The organic light-emitting structure **330** may have an anode connected to the output electrode of the driving transistor **310** and a cathode receiving a common voltage ELVSS. The organic light-emitting structure **330** may emit light based on the output current I_D from the driving transistor **310** to display an image.

(20) Referring to FIG. **1** again, the image may not be displayed in the peripheral region **120** of the organic light-emitting display device **100**. The peripheral region **120** may substantially surround the display region **110**, and may be located in outer portion of the organic light-emitting display device **100**.

(21) The embedded circuit part **180** having a driver circuit for driving the pixels PX may be disposed in the peripheral region **120**. The embedded circuit part **180** may include the driver circuit (e.g., a gate driver and/or a light emission control driver) for driving the pixels PX, and may further include an inspection circuit, etc. The embedded circuit part **180** may be located in both sides of the display region **110**, however, the embedded circuit part **180** may be located in one side of the display region **110**, alternatively.

(22) The pad part **170** may be disposed in the peripheral region **120**. Various signals (e.g., the data signal) may be applied to the pad part **170** from an external device, and a flexible printed circuit **197**, on which integrated circuits are mounted, may be electrically connected to the pad part **170**.

(23) The first wiring part **130** and the second wiring part **140** may be disposed in the peripheral region **120**. The first wiring part **130** may be located at one side of the organic light-emitting display device **100**. The second wiring part **140** may be located to surround the display region **110**.

(24) One side of the first wiring part **130** may be connected to the flexible printed circuit **197**. Another side of the first wiring part **130** may be connected to the power voltage lines **195**. The first wiring part **130** may transfer the driving voltage ELVDD from the flexible printed circuit **197** to the power voltage lines **195**.

(25) The second wiring part **140** may be connected to the flexible printed circuit **197**. The second wiring part **140** may transfer the common voltage ELVSS from the flexible printed circuit **197** to the pixels PX.

(26) The encapsulation layer **150** may be, e.g., continuously, disposed over the pixels PX, over the first wiring part **130**, and over the second wiring part **140**. The encapsulation layer **150** may block vapor or oxygen from outside to protect the pixels PX, the first wiring part **130**, and the second wiring part **140**.

(27) In example embodiments, the encapsulation layer **150** may include at least one organic layer and at least one inorganic layer. For example, the encapsulation layer **150** may include at least two inorganic layers and at least one organic layer. In this case, the organic layer may be disposed

between the inorganic layers. In other words, the encapsulation layer **150** may have a structure in which a first inorganic layer **151** in FIG. **4**, an organic layer **152** in FIG. **4**, and a second inorganic layer **153** in FIG. **4** are sequentially stacked. For example, the inorganic layer may include at least one of aluminum oxide (AlO.sub.x), silicon oxide (SiO.sub.x), silicon nitride (SiN.sub.x), silicon oxynitride (SiO.sub.xN.sub.y), silicon carbide (SiC.sub.x), titanium oxide (TiO.sub.x), zirconium oxide (ZrO.sub.x) and/or zinc oxide (ZnO.sub.x). Furthermore, the organic layer may include epoxy, polyimide (PI), polyethylene terephthalate (PET), polycarbonate (PC), polyethylene (PE) and/or polyacrylate. The inorganic layer(s) may block permeation of oxygen or vapor, and the organic layer may provide flexibility to the organic light-emitting display device **100**.

(28) FIG. **3** is a plan view illustrating an enlarged portion 'X' of the organic light-emitting display device in FIG. **1**.

(29) Referring to FIGS. **1** and **3**, the peripheral region **120** may include a first sub-region **122**, a second sub-region **124**, and a third sub-region **126**. The first sub-region **122** may be located adjacent to the display region **110**. The second sub-region **124** may be located adjacent to the flexible printed circuit **197**, e.g., most remote from the display region **110** among the first through third sub-regions **122** through **126**. The third sub-region **126** may be located between the first sub-region **122** and the second sub-region **124**.

(30) In example embodiments, organic insulation layers (e.g., an insulation layer and a pixel defining layer) may be disposed between the first and second wiring parts **130** and **140**, and the encapsulation layer **150**. When the organic insulation layers are disposed in the entirety of the display region **110** and the peripheral region **120**, vapor or oxygen may permeate from outside to the peripheral region **120** to pass through the organic insulation layer into the display region **110**. In this case, the vapor or oxygen may contact the pixels PX disposed in the display region **110**, so that the organic light-emitting structure **330** may be degraded.

(31) Therefore, according to embodiments, a portion of the organic insulation layer in the third sub-region **126** of the peripheral region **120** may be removed to prevent above mentioned degradation of the organic light-emitting structure **330**. Therefore, an inflow path of the vapor or oxygen from outside to the organic light-emitting structure **330** may be blocked, so that the degradation of the organic light-emitting structure **330** may be prevented. This will be described in more detail below with reference to FIG. **4**.

(32) FIG. **4** is a cross-sectional view along line I-I' in FIG. **1**. For example, the right side of FIG. **4** illustrates the horizontal part of line I-I' (crossing elements **130** and **140** in the peripheral region **120**), and the left side of FIG. **4** illustrates the vertical part of line I-I' (crossing the pixel PX in the display region **110**).

(33) Referring to FIG. **4**, the organic light-emitting display device **100** may include a substrate **200**, a pixel circuit, the first wiring part **130**, the second wiring part **140**, a compensation layer **160**, the organic light-emitting structure **330**, the encapsulation layer **150**, etc. In example embodiments, the first wiring part **130** may include a first wiring **132**, and the second wiring part **140** may include a second wiring **142**. The organic light-emitting structure **330** may include a pixel electrode **260**, an organic light-emitting layer **280**, and a common electrode **290**, etc.

(34) The pixel circuit may include the driving transistor **310** and the capacitor **320**. The pixel circuit may further include the switching transistor of FIG. **2** (not illustrated in FIG. **4** for convenience). For example, each of the driving transistor **310** and the switching transistor may correspond to a thin film transistor. The driving transistor **310** may include a first active pattern **210a**, a first gate electrode **220a**, a source electrode **240**, and a drain electrode **245**. The capacitor **320** may include a second active pattern **210b** and a second gate electrode **220b**.

(35) For example, the substrate **200** may include a transparent substrate, e.g., a glass substrate, a quartz substrate, a transparent plastic substrate, etc. In another example, the substrate **200** may include a flexible substrate.

(36) A buffer layer **205** may be disposed on the substrate **200**. The buffer layer **205** may extend

from the display region **110** to the peripheral region **120**. The buffer layer **205** may prevent a diffusion of impurities generated from the substrate **200**, and may control a transfer rate of heat during a crystallization process for forming the first and second active patterns **210a** and **210b**. For example, the buffer layer **205** may include silicon oxide (SiO.sub.x), silicon nitride (SiN.sub.x), silicon oxynitride (SiO.sub.xN.sub.y), etc. Here, the buffer layer **205** may have a single layer structure or a multi-layer structure. Alternatively, the buffer layer **205** may be omitted.

(37) The first active pattern **210a** and the second active pattern **210b** may be disposed on the buffer layer **205**. Each of the first and second active patterns **210a** and **210b** may include a silicon compound or a polysilicon. A source region and a drain region may be disposed at both sides of the first active pattern **210a**, and each of the source region and the drain region may include impurities. In addition, the second active pattern **210b** may include impurities, and the impurities may be distributed around the second active pattern **210b**. In some example embodiments, each of the first and second active patterns **210a** and **210b** may include oxide semiconductor, e.g., indium-gallium-zinc oxide (IGZO), zinc-tin oxide (ZTO) or indium-tin-zinc oxide (ITZO).

(38) A gate insulation layer **215** may be disposed on the buffer layer **205** to substantially cover the first and second active patterns **210a** and **210b**. The gate insulation layer **215** may extend from the display region **110** to the peripheral region **120**. For example, the gate insulation layer **215** may include silicon oxide (SiO.sub.x), silicon nitride (SiN.sub.x), silicon oxynitride (SiO.sub.xN.sub.y), etc. Here, the gate insulation layer **215** may have a single layer structure or a multi-layer structure.

(39) The first gate electrode **220a** and the second gate electrode **220b** may be disposed on the gate insulation layer **215** in the display region **110**. The first gate electrode **220a** and the second gate electrode **220b** may be disposed over the first active pattern **210a** and the second active pattern **210b**, respectively. For example, each of the first and second gate electrodes **220a** and **220b** may include aluminum (Al), silver (Ag), tungsten (W), copper (Cu), nickel (Ni), chromium (Cr), molybdenum (Mo), titanium (Ti), platinum (Pt), tantalum (Ta), neodymium (Nd), scandium (Sc), alloys thereof, or nitrides thereof. These may be used alone or in a combination thereof.

(40) The first wiring **132** and the second wiring **142** may be disposed on the gate insulation layer **215** in the peripheral region **120**. The first wiring **132** and the second wiring **142** may be spaced apart from each other at a predetermined distance, and may be disposed at substantially the same level over the substrate **200**, e.g., at a same distance from a bottom surface of the substrate **200**. Moreover, the first and second wirings **132** and **142** may be disposed at substantially the same level as the first and second gate electrodes **220a** and **220b** over the substrate **200**, e.g., the first and second wirings **132** and **142** may be directly on a same element as the first and second gate electrodes **220a** and **220b**. In example embodiments, the first and second wirings **132** and **142** may include substantially the same material as the first and second gate electrodes **220a** and **220b**.

(41) An insulation interlayer **235** may be disposed on the gate insulation layer **215** in the display region **110** to substantially cover the first and second gate electrodes **220a** and **220b**. For example, the insulation interlayer **235** may include silicon oxide (SiO.sub.x), silicon nitride (SiN.sub.x), silicon oxynitride (SiO.sub.xN.sub.y), etc. Here, the insulation interlayer **235** may have a single layer structure or a multi-layer structure.

(42) The source electrode **240** and the drain electrode **245** may be disposed on the insulation interlayer **235**. Each of the source electrode **240** and the drain electrode **245** may pass through the insulation interlayer **235** and the gate insulation layer **215** to contact the first active pattern **210a**. For example, each of the source electrode **240** and the drain electrode **245** may include aluminum (Al), silver (Ag), tungsten (W), copper (Cu), nickel (Ni), chromium (Cr), molybdenum (Mo), titanium (Ti), platinum (Pt), tantalum (Ta), neodymium (Nd), scandium (Sc), alloys thereof, or nitrides thereof. These may be used alone or in a combination thereof.

(43) In example embodiments, the driving transistor may include the first active pattern **210a**, the gate insulation layer **215**, the first gate electrode **220a**, the source electrode **240**, and the drain electrode **245**, and the capacitor **320** may include the second active pattern **210b**, the gate insulation

layer **215**, and the second gate electrode **220b**.

(44) An insulation layer **250** may be disposed on the insulation interlayer **235**. The insulation layer **250** may substantially cover the source electrode **240** and the drain electrode **245**. A via structure for electrically connecting the pixel electrode **260** and the drain electrode **245** may be provided in the insulation layer **250**. In example embodiments, a contact hole exposing a portion of the drain electrode **245** may be provided in the insulation layer **250**, and the pixel electrode **260** may fill the contact hole to contact the drain electrode **245**. In other example embodiments, a contact including a conductive material may be provided in the contact hole, and the pixel electrode **260** and the drain electrode **245** may contact a top portion and a bottom portion of the contact, respectively. The insulation layer **250** may substantially serve as a planarization layer for upper structures. For example, the insulation layer **250** may include an organic material, e.g., polyimide, epoxy-based resin, acryl-based resin, polyester, or the like.

(45) The insulation layer **250** may be disposed in the display region **110**. However, the insulation layer **250** may not be disposed in the third sub-region **126** of FIG. 3. Therefore, a path for vapor or oxygen to move from outside to elements in the display region **110** may be blocked.

(46) The pixel electrode **260** may be disposed on the insulation layer **250**. As discussed previously, the pixel electrode **260** may fill the contact hole to contact the drain electrode **245**, or the pixel electrode **260** may be electrically connected to the drain electrode **245** through the contact provided in the contact hole. For example, the pixel electrode **260** may include aluminum (Al), silver (Ag), tungsten (W), copper (Cu), nickel (Ni), chromium (Cr), molybdenum (Mo), titanium (Ti), platinum (Pt), tantalum (Ta), neodymium (Nd), scandium (Sc), or alloys thereof. Alternatively, the pixel electrode **260** may include a transparent conductive material. In example embodiments, the pixel electrode **260** may be individually disposed per each pixel.

(47) A pixel defining layer **270** may be disposed in the display region **110**, and may not be disposed in the third sub-region **126**. Therefore, a path for vapor or oxygen to move from outside to elements in the display region **110** may be blocked.

(48) The organic light-emitting layer **280** may be disposed on the pixel electrode **260**. The organic light-emitting layer **280** may include a host material excited by holes and electrons, and a dopant material for facilitating an absorbance and a release of energy and improving a light-emitting efficiency. Alternatively, the organic light-emitting structure **330** may further include a hole transport layer (HTL) interposed between the pixel electrode **260** and the organic light-emitting layer **280**. The organic light-emitting structure **330** may further include an electron transport layer (ETL) interposed between the common electrode **290** and the organic light-emitting layer **280**.

(49) The common electrode **290** may be disposed on the organic light-emitting layer **280** and on the pixel defining layer **270**. The common electrode **290** may face the pixel electrode **260** with respect to the organic light-emitting layer **280**. For example, the common electrode **290** may include a metal having a relatively low work function, e.g., aluminum (Al), silver (Ag), tungsten (W), copper (Cu), nickel (Ni), chromium (Cr), molybdenum (Mo), titanium (Ti), platinum (Pt), tantalum (Ta), neodymium (Nd), scandium (Sc), or alloys thereof. In example embodiments, the pixel electrode **260** and the common electrode **290** may be provided as an anode and a cathode of the organic light-emitting structure **330**, respectively.

(50) The encapsulation layer **150** may substantially cover an entirety of the display region **110** and the peripheral region **120** over the substrate **200**. In example embodiments, the encapsulation layer **150** may be disposed on the common electrode **290** in the display region **110**, and may be disposed over the first and second wirings **132** and **142** in the peripheral region **120**. Here, as illustrated in a portion 'Y' in FIG. 4, the encapsulation layer **150** may have a stepped region along a profile of the first and second wirings **132** and **142**. If the encapsulation layer **150** were to be directly located on a top surface and a sidewall of each of the first and second wirings **132** and **142**, damage (e.g., a crack) could occur in the stepped region of the encapsulation layer **150**, so that properties of the organic light-emitting display device **100** could be decreased.

(51) Therefore, according to embodiments, the compensation layer **160** is disposed between the first and second wirings **132** and **142**, and the encapsulation layer **150**, and may surround the top surface and the sidewall of each of the first and second wirings **132** and **142**. In other words, the compensation layer **160** may, e.g., conformally, cover the top surface and the sidewall of each of the first and second wirings **132** and **142**, e.g., to reduce a height of a stepped region formed by the first and second wirings **132** and **142**. Therefore, the compensation layer **160** may compensate a stepped region derived from the first and second wirings **132** and **142**, so damage (e.g., a crack) on the encapsulation layer **150** may be prevented.

(52) In detail, the compensation layer **160** may be disposed on the gate insulation layer **215** in the peripheral region **120**. In example embodiments, the compensation layer **160** and the insulation interlayer **235** may be disposed at substantially the same level over the substrate **200**. In other words, the insulation interlayer **235** may extend, e.g., continuously, from the display region **110** to the peripheral region **120**, and may serve as the compensation layer **160** on the top surface and the sidewall of each of the first and second wirings **132** and **142** in the peripheral region **120**. Here, the compensation layer **160** may include substantially the same material as the insulation interlayer **235**, e.g., the compensation layer **160** and the insulation interlayer **235** may be parts of a same layer. The compensation layer **160** may be disposed between the first and second wirings **132** and **142**, and the encapsulation layer **150**, so that the compensation layer **160** may compensate the stepped region derived from the first and second wirings **132** and **142**, and damage (e.g., a crack) on the encapsulation layer **150** may be prevented.

(53) FIG. 5 is a cross-sectional view illustrating a cross-sectional view taken along line I-I' in FIG. 1 in accordance with other embodiments.

(54) Referring to FIG. 5, an organic light-emitting display device **100'** may include the substrate **200**, the pixel circuit, the first wiring part **130**, the second wiring part **140**, the compensation layer **160**, the organic light-emitting structure **330**, the encapsulation layer **150**, etc. In example embodiments, the first wiring part **130** may include the first wiring **132** and a third wiring **134**, and the second wiring part **140** may include the second wiring **142** and a fourth wiring **144**. The organic light-emitting structure **330** may include the pixel electrode **260**, the organic light-emitting layer **280** and the common electrode **290**, etc. Detailed description of elements in FIG. 5 which are substantially the same as or similar to those described previously with reference to FIG. 4 will not be repeated.

(55) The pixel circuit may include the driving transistor **310** and the capacitor **320**. The pixel circuit may further include the switching transistor, however, the switching transistor is not illustrated in FIG. 5 for convenience of illustration. For example, each of the driving transistor **310** and the switching transistor may correspond to a thin film transistor. The driving transistor **310** may include an active pattern **210**, a second gate electrode **230a**, the source electrode **240**, and the drain electrode **245**. The capacitor **320** may include a first gate electrode **220** and a third gate electrode **230b**.

(56) The first gate electrode **220** may be disposed on the gate insulation **215** in the display region **110**. For example, the first gate electrode **220** may include aluminum (Al), silver (Ag), tungsten (W), copper (Cu), nickel (Ni), chromium (Cr), molybdenum (Mo), titanium (Ti), platinum (Pt), tantalum (Ta), neodymium (Nd), scandium (Sc), alloys thereof, or nitrides thereof. These may be used alone or in a combination thereof.

(57) The first wiring **132** and the second wiring **142** may be disposed on the gate insulation layer **215** in the peripheral region **120**. The first wiring **132** and the second wiring **142** may be spaced apart from each other at a predetermined distance, and may be disposed at substantially the same level over the substrate **200**. Moreover, the first and second wirings **132** and **142** may be disposed at substantially the same level as the first gate electrode **220** over the substrate **200**. In example embodiments, the first and second wirings **132** and **142** may include substantially the same material as the first gate electrode **220**.

(58) A second gate insulation layer **225** may be disposed on the first gate insulation layer **215** in the display region **110** to substantially cover the first gate electrode **220**. The second gate insulation layer **225** may insulate the first gate electrode **220** and the third gate electrode **230b** which are electrodes of the capacitor **320**. For example, the second gate insulation layer **225** may include silicon oxide (SiO.sub.x), silicon nitride (SiN.sub.x), silicon oxynitride (SiO.sub.xN.sub.y), etc. Here, the second gate insulation layer **225** may have a single layer structure or a multi-layer structure.

(59) An interwiring insulation layer **190** may be disposed on the first gate insulation layer **215** in the peripheral region **120** to substantially surround a top surface and a sidewall of each of the first and second wirings **132** and **142**. In example embodiments, the second gate insulation layer **225** and the interwiring insulation layer **190** may be disposed at substantially the same level over the substrate **200**. In other words, the second gate insulation layer **225** may extend from the display region **110** to the peripheral region **120**, and may serve as the interwiring insulation layer **190** substantially surrounding the top surface and the sidewall of each of the first and second wirings **132** and **142** in the peripheral region **120**. Here, the interwiring insulation layer **190** may include substantially the same material as the second gate insulation layer **225**, e.g., the interwiring insulation layer **190** and the second gate insulation layer **225** may be parts of a same layer.

(60) The second gate electrode **230a** and the third gate electrode **230b** may be disposed on the second gate insulation layer **215**. The second gate electrode **230a** and the third gate electrode **230b** may be disposed over the active pattern **210** and the first gate electrode **220**, respectively. For example, each of the second and third gate electrodes **230a** and **230b** may include aluminum (Al), silver (Ag), tungsten (W), copper (Cu), nickel (Ni), chromium (Cr), molybdenum (Mo), titanium (Ti), platinum (Pt), tantalum (Ta), neodymium (Nd), scandium (Sc), alloys thereof, or nitrides thereof. These may be used alone or in a combination thereof.

(61) In example embodiments, the third wiring **134** and the fourth wiring **144** may be disposed on the interwiring insulation layer **190**. The third wiring **134** and the fourth wiring **144** may substantially overlap the first wiring **132** and the second wiring **142**, respectively. The third wiring **134** and the fourth wiring **144** may be spaced apart from each other at a predetermined distance, and may be disposed at substantially the same level over the substrate **200**. Moreover, the third and fourth wirings **134** and **144** may be disposed at substantially the same level as the second and third gate electrodes **230a** and **230b** over the substrate **200**. In example embodiments, the third and fourth wirings **134** and **144** may include substantially the same material as the second and third gate electrodes **230a** and **230b**.

(62) The third wiring **134** may be electrically connected to the first wiring **132**, and the fourth wiring **144** may be electrically connected to the second wiring **142**. In example embodiments, the third and fourth wirings **134** and **144** may contact the first and second wirings **132** and **142**, respectively, through contact holes **172** and **174**. More specifically, a first contact hole **172** exposing the first wiring **132** and a second contact hole **174** exposing the second wiring **142** may be provided in the interwiring insulation layer **190**. The third wiring **134** may fill the first contact hole **172** to contact the first wiring **132**, and the fourth wiring **144** may fill the second contact hole **174** to contact the second wiring **142**. Therefore, a resistance of each of the first wiring part **130** and the second wiring part **140** may be decreased.

(63) The compensation layer **160** may be disposed between the third and fourth wirings **134** and **144**, and the encapsulation layer **150**, and may surround the top surface and the sidewall of each of the third and fourth wirings **134** and **144**. The compensation layer **160** may be disposed on the interwiring insulation layer **190** in the peripheral region **120**. In example embodiments, the compensation layer **160** and the insulation interlayer **235** may be disposed at substantially the same level over the substrate **200**. In other words, the insulation interlayer **235** may extend from the display region **110** to the peripheral region **120**, and may serve as the compensation layer **160** substantially surrounding the top surface and the sidewall of each of the third and fourth wirings

134 and **144** in the peripheral region **120**. The compensation layer **160** may compensate the stepped region derived from the third and fourth wirings **134** and **144**. Here, the compensation layer **160** may include substantially the same material as the insulation interlayer **235**.

(64) The compensation layer **160** may be disposed between the third and fourth wirings **134** and **144**, and the encapsulation layer **150**, so that the compensation layer **160** may compensate the stepped region derived from the third and fourth wirings **134** and **144** as illustrated in a portion 'Z' in FIG. 5, and damage (e.g., a crack) on the encapsulation layer **150** may be prevented.

(65) The organic light-emitting display device according to example embodiments may be applied to various electronic devices. For example, the organic light-emitting display devices may be applied to computers, notebooks, mobile phones, smart phones, smart pads, personal media players, personal digital assistants, MP3 players, digital cameras, video camcorders, etc.

(66) By way of summation and review, power voltage wirings including metal may be located in a peripheral region of the organic light-emitting display device. When the encapsulation layer is disposed on and directly contacts the power voltage wirings, the encapsulation layer may be damaged or cracked due to stress generated from the power voltage wirings.

(67) In contrast, example embodiments provide an organic light-emitting display device including a compensation layer that compensates for a stepped region derived from wirings. That is, according to example embodiments, the organic light-emitting display device may include the compensation layer disposed between the first and second wirings, and the encapsulation layer, so that the stepped region derived from the first and second wirings may be substantially compensated, and damage (e.g., crack) to the encapsulation layer may be prevented. Furthermore, the organic light-emitting display device may further include the third and fourth wirings electrically connected to the first and second wirings, respectively, such that electrical resistance of the first and second wirings may be decreased, and electrical properties of the organic light-emitting display device may be improved.

(68) Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

Claims

1. A light-emitting display device, comprising: a substrate having a display region and a peripheral region; a plurality of pixels on the substrate in the display region; a first wiring on the substrate in the peripheral region; a first insulation layer partially covering a top surface and a sidewall of the first wiring; a second wiring on the first insulation layer, the second wiring being electrically connected to the first wiring; a second insulation layer on the second wiring, the second insulation layer covering a top surface and a sidewall of the second wiring; and an encapsulation layer including a plurality of stacked layers, the encapsulation layer encapsulating the plurality of pixels and extending from the display region to the peripheral region on the second insulation layer, wherein the second insulation layer is disposed between the second wiring and a lowermost inorganic layer of the plurality of stacked layers, and a first portion of the second insulation layer contacts the first insulation layer.
2. The light-emitting display device as claimed in claim 1, wherein the first insulation layer is formed along a profile of the first wiring.

3. The light-emitting display device as claimed in claim 1, wherein the second insulation layer is formed along a profile of the second wiring.
 4. The light-emitting display device as claimed in claim 1, further comprising a conductive layer on the second insulation layer.
 5. The light-emitting display device as claimed in claim 4, wherein a second portion of the second insulation layer contacts the conductive layer.
 6. The light-emitting display device as claimed in claim 4, further comprising a planarization layer disposed in the display region and on the conductive layer, the planarization layer being absent from a portion of the peripheral region adjacent to the display region.
 7. The light-emitting display device as claimed in claim 1, wherein the encapsulation layer is at least partially directly on the second insulation layer in the peripheral region.
 8. The light-emitting display device as claimed in claim 1, wherein the second insulation layer has a partially stepped shape in a cross-sectional view.
 9. The light-emitting display device as claimed in claim 1, wherein the first insulation layer has a first contact hole exposing the first wiring, and the second wiring fills the first contact hole to contact the first wiring.
 10. The light-emitting display device as claimed in claim 1, wherein each of the plurality of pixels includes: a pixel circuit having an active pattern, a first gate electrode, a gate insulation layer, a second gate electrode, the second insulation layer, a source electrode, and a drain electrode; and an organic light-emitting structure having a pixel electrode, an organic light-emitting layer, and a common electrode, the organic light-emitting structure being electrically connected to the pixel circuit.
 11. The light-emitting display device as claimed in claim 10, wherein the first wiring and the first gate electrode are at a same level over the substrate.
 12. The light-emitting display device as claimed in claim 10, wherein the second wiring and the second gate electrode are at a same level over the substrate.
 13. The light-emitting display device as claimed in claim 10, wherein the first insulation layer and the gate insulation layer are at a same level over the substrate.
 14. The light-emitting display device as claimed in claim 10, further comprising a conductive layer on the second insulation layer, wherein the conductive layer includes the source electrode and the drain electrode.
 15. The light-emitting display device as claimed in claim 1, wherein a portion of the second wiring is between the first insulation layer and the second insulation layer in a horizontal direction.
 16. The light-emitting display device as claimed in claim 1, wherein the second insulation layer prevents a crack from occurring in the encapsulation layer due a stepped region caused by the first wiring and the second wiring.
 17. The light-emitting display device as claimed in claim 1, wherein the second wiring overlaps the first wiring in a plan view.
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